

Peter Massih

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EDUCATION

EPFL, MSc in Computer Science (Data Analytics and Machine Learning), **GPA 5.5/6.0** 2023 – 2026
ETH Zurich, exchange year in Computer Science and Electrical Engineering 2022 – 2023
EPFL, BSc in Communication Systems (Electrical Engineering and Computer Science) 2020 – 2023
Coursework: Machine Learning, Modern NLP, Statistical Learning, Optimization, Probability and Statistics, Advanced Algorithms

EXPERIENCE

NEC Laboratories America Princeton, NJ
Machine Learning Research Intern Feb 2025 – Oct 2025, May 2026 – Aug 2026

- Built SQuID, a 2,000-question satellite-image benchmark (first-author arXiv preprint), and raised GPT-5's accuracy on it **from 28.1% to 42.0%** by having it write sandboxed Python that measures segmentation masks.
- Trained the ConvNeXt and DINOv3-Mask2Former models behind these masks, shown at NEC's Open House.
- Lifted Qwen2.5-VL-7B accuracy on SQuID **from 16.1% to 40.7%** with LoRA on 75,687 training pairs.
- Showed in a first-author ICLR 2027 submission that a VLM's latent "visual thoughts" (LVR) add **14.2 points** in counting and 12.9 in left-or-right accuracy when the image is removed.
- Designed a VLM that reasons with one token per object, each decoded into a mask under a Hungarian-matching loss (first-author paper in preparation), and post-trained mixture-of-experts VLMs with SFT and RL (GRPO).

Bank Lombard Odier & Co Ltd Geneva, Switzerland
Software Engineer Intern, AI/ML (EPFL Master's thesis, advised by Prof. Amir Zamir) Oct 2025 – Apr 2026

- Deployed **3 AI systems** used daily by bankers, fund screeners and tax specialists, under FINMA banking rules.
- Cut indexing of a 52-page prospectus **from 17 minutes to 44 seconds** in the production RAG system, with 95% of answers citing the right page, by speeding up embedding inference (ONNX Runtime, llama.cpp, Azure).
- Built an on-premises LLM pipeline (Qwen3-30B-A3B) for buy and sell recommendations, **96%** of which matched the bank's equity ratings, CIO views and client holdings, on par with GPT-5.1 with RAG (93%).

Ageospatial, geospatial AI startup Lausanne, Switzerland
Full-Stack AI Engineer Intern Jul 2024 – Oct 2024

- Developed a LangGraph map agent for geospatial questions, adopted by **3 Swiss government agencies**.

PUBLICATIONS

- P. A. Massih**, E. Cosatto. Reasoning with Pixel-level Precision: QVLM Architecture and SQuID Dataset for Quantitative Geospatial Analytics. *Preprint, arXiv:2601.13401, 2026*. Dataset on Hugging Face.
- P. A. Massih**, E. Cosatto. What Is a Visual Thought Worth? When Continuous Visual States Help Vision-Language Models Reason. *Under review at ICLR 2027*.
- S. Lu, **P. A. Massih**, E. Cosatto, M. R. Min. Test-Time Reasoning with Computational Evidence for Spatial Question Answering. *Under review at ICLR 2027*.

PROJECTS

- 3D-BlockGen**. Turned text into buildable LEGO models with a two-stage diffusion model (shape, then color) trained on a **542,292**-sample voxelized Objaverse dataset, beating single-stage variants on IoU, F1 and color.
- LLM Teaching Assistant**. Fine-tuned Llama-3-8B with SFT and DPO, raising how often it prefers the better answer on about 800 held-out course questions **from 55% to 63%**, then shrank it 5.8x with 4-bit GPTQ.
- Reasoning-Model Paraphraser**. Fine-tuned DeepSeek-R1-Distill-Qwen-7B with LoRA to write paraphrases **twice as diverse** as BART's (BLEU diversity 0.51 vs 0.23) with similar meaning (BERTScore 0.95 vs 0.96).
- RL on Mountain Car**. Shortened DQN training **from 300 to 200 episodes** with a direction-aware reward.

SKILLS

Research areas: large language models (LLMs), vision-language models, deep learning, post-training, evaluation
Machine learning: PyTorch, distributed training (FSDP), Hugging Face, SFT, DPO, GRPO, LoRA, vLLM, RAG
Programming and systems: Python, C++, C, Java, SQL, Slurm, Docker, Kubernetes, Linux, Git
Languages: English, French and Arabic (native), German (elementary)

TEACHING AND ACTIVITIES

EPFL Teaching Assistant, Java (about 600 students) and Computer Systems in C (about 300) Sep 2023 – Jan 2025
Optiver Career Kickstarter, built a Python market-making strategy in a 5-day trading program Apr 2025
Lebanese Swimming Federation, national silver in butterfly and bronze in freestyle 2016 – 2021